Rugby Taylor

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EDUCATION

Brigham Young University, Ira A. Fulton College of Engineering Apr 2027
Bachelor of Science, Electrical Engineering
- **Focus:** Power Electronics, Energy Storage, Hardware Testing
- **Certifications:** FE Electrical and Computer Exam (In Progress), PII Q-LS UPS Service Technician

TECHNICAL SKILLS

Power & Energy Systems: UPS systems (60-120 kVA), lithium battery modules (50V), HV interlocks
Hardware Design: KiCad (schematics, PCB layout, flex PCB/FPC), analog circuit design, LTspice
HV Safety: LOTO, arc flash, isolation procedures, HV shutdown system design, system-level fault diagnosis
Embedded & Controls: C/C++, Python, MATLAB, ESP32, STM32, PLC interface (Beckhoff TwinCAT, EtherCAT), real-time measurement systems
Fabrication & Prototyping: Soldering, custom hardware assembly, milling/machining, welding

RELEVANT EXPERIENCE

Nusano (Particle Accelerator) May 2026 - Aug 2026
High Power Systems Intern
- Integrated a precision sample-and-hold circuit enabling PLCs to capture high-intensity arc-flash signals previously beyond sampling limits, validated to <0.5% total system error
- Architected a multi-input, LOTO-compliant keyed interlock system integrating 8 HV safety signals into a single plug-and-play enclosure, replacing fragmented field wiring

BYU Racing Team (FSAE Electric Car) Jan 2026 - Present
Chief Electrical Systems Officer
- Spearheaded a full redesign of the vehicle shutdown architecture integrating, precharge, discharge, BSPD, IMD/BMS resets, and ready-to-drive indicator logic
- Directed full BMS PCB designs for a 108S4P tractive battery, independently developing custom FPC voltage-sensing hardware validated in an industry design review
- Leading development of a custom 3-phase tractive inverter (600V, 150A)
- Designed analog precharge control circuitry establishing the team's first fully functional precharge system

Rescue Power (UPS Systems) Feb 2025 - Present
Field Engineer
- Executed installation and acceptance testing of 60–120 kVA UPS systems supporting mission-critical loads
- Refurbished 58+ high-capacity battery modules, extending system service life and improving reliability
- Diagnosed and resolved high-urgency electrical failures under time-critical conditions, restoring full UPS functionality and minimizing downtime

PERSONAL PROJECTS

Automated Signal Monitoring & Control System July 2025 - Present
- Developed an autonomous Python-based signal-analysis pipeline evaluating high-frequency data at 1Hz against multi-month historical baselines
- Designed for fully autonomous 24/7 execution and monitoring without manual intervention

Embedded Digital Oscilloscope Oct 2025 - Jan 2026
- Engineered a real-time oscilloscope platform on ESP32 with custom analog front-end and LUT-based calibration achieving <2% error with fully functioning firmware builds and CI/CD pipeline

Energy Storage System Design & Integration Aug 2023 - Jan 2024
- Designed and assembled a 10S2P, 150A peak lithium pack, reducing system cost by ~12% while maintaining thermal and electrical safety margins